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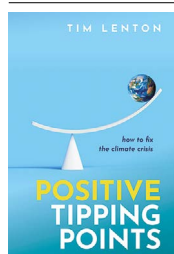
Indigenous people in Brazil and elsewhere are pressing for a stronger voice in climate negotiations.

How collective action can prevent catastrophic climate change

Even small acts can make a big difference in driving positive environmental impacts. **By Michael Jakob**

Earth-systems scientist Tim Lenton has long been emphasizing the importance of ‘tipping points’ for climate change. In a seminal paper published in 2008, he and his co-authors identified several elements of Earth’s climate system that might shift to a different state irreversibly once physical thresholds are crossed (T. M. Lenton *et al.* *Proc. Natl Acad. Sci. USA* **105**, 1786–1793; 2008). For example, the Greenland ice sheet, the Amazon rainforest and circulation patterns in the Atlantic Ocean could change suddenly as the world warms.

Breaches of climate tipping points are a reason for great concern and warrant rapid reductions in greenhouse-gas emissions. Yet



**Positive Tipping Points:
How to Fix the Climate
Crisis**

Tim Lenton
Oxford Univ. Press (2025)

some types of tipping point could instead help to slow down climate change. Social ones, for example, could accelerate the transition to climate neutrality. Policies that have helped to reduce the cost of solar panels, wind turbines and batteries, for example, have encouraged consumers and companies to use electric and renewable-energy sources. These trends are unlikely to be reversed even if such policies are revoked.

In *Positive Tipping Points*, Lenton takes a closer look at these social forces and discusses how they can be harnessed to avoid

Books & arts

further drastic climate change. His engaging and thought-provoking discussion of how individuals can bring about positive tipping points and contribute to climate action is based on state-of-the-art research and packed with useful facts and figures. It is a breath of fresh air at a time when people are increasingly prone to feelings of ‘climate doom’, the idea that a climate catastrophe is unavoidable.

Big changes

The first part of the book highlights that changes in the climate system and society are often non-linear and can occur rapidly. For systems with self-reinforcing feedback loops, a small perturbation might be enough to tip the climate system from one equilibrium state to another. For example, higher temperatures accelerate the melting of sea ice, reducing the amount of solar radiation that is reflected back into space, thus triggering further warming.

Past tipping events include the transition, roughly 12,000 years ago, from the last ice age to the current Holocene interglacial period. Such drastic changes in environmental conditions had implications for human societies, too. The end of that big freeze enabled people to switch from foraging to farming as their main means of food production.

Other wholesale cultural shifts might be expected in a warmer world. If humans are forced to live outside the climatic niche in which our societies evolved, social stability might be threatened. But other positive tipping points could emerge that work in favour of climate-change mitigation.

The second part of Lenton's book discusses

how positive tipping points can help to avert a climate catastrophe. He includes numerous examples of how actions taken by small groups of people have resulted in large-scale social change. For example, Lenton describes how his aunt Lilian's fight for women's suffrage helped to change social norms, and how a small number of climate activists, such as Greta Thunberg, have sparked a global movement.

The author argues that the key to pushing social change towards a climate-friendly

“Citizens’ assemblies can help governments to find solutions that meet communities’ needs.”

economy lies in feedback loops – progress in one area increases the probability of improvements in others. Social activism, alternative forms of democracy, technological progress and government policies could reinforce each other, allowing societies to overcome the current state of ‘carbon lock-in’ by reducing their reliance on fossil fuels.

Small steps

Citizens’ assemblies can help governments to find solutions that meet communities’ needs, such as extra charging infrastructure for electric vehicles, subsidies for household heat pumps, input in urban planning and improvements to public transport. Some cities, such as Porto Alegre in Brazil, use participatory budgeting processes that give citizens a say

in how their taxes are spent, increasing trust in policymakers. Using a similar approach, Climate Assembly UK, a representative group of 108 people from across the country, met in 2020 to discuss how to achieve the government's target of reaching ‘net zero’ carbon emissions by 2050. It recommended that the government pursue more ambitious climate policies, including measures to reduce people's consumption of animal products and to transition to carbon-neutral heating.

Lenton gives examples of policies that have driven transformative change. Carbon pricing – when a monetary cost is applied to emissions – has shifted power generation away from coal in many countries, including in the United Kingdom. Mandates to phase out the use of fossil fuels, such as Chile's commitment to stop coal use by 2040, provide a clear long-term perspective for investors. Governments can help innovative technologies to achieve market readiness through direct investment or public procurement. Financial support for clean energy, such as feed-in tariffs that guarantee the market price of electricity for energy producers, has brought down costs and accelerated the deployment of renewable-energy technologies globally.

Lenton discusses how to spark transformation by deliberately bringing about ‘cascades’ – when one positive tipping point triggers further ones. For example, green-hydrogen fuel, produced with renewable-energy sources, can be used to lower emissions in sectors where these are hard to abate, such as steelmaking, shipping and aviation. Adopting such clean fuels in one sector will reduce the cost of producing green hydrogen owing to economies of scale, enabling similar shifts in other sectors.

It thus seems worthwhile, Lenton argues, to focus decarbonization efforts on factors that will have a positive effect across several sectors and systems, and energy is one such target. But it's hard to spot all the areas that might have big knock-on effects. That's why it's important to think outside the box, try out many forms of social organization and trial innovative policy approaches. Who knows, some of these experiments might just mobilize unexpected support for a transition to climate neutrality.

Positive Tipping Points shows that addressing climate change need not necessarily entail personal sacrifice yet can still be meaningful on a deeply personal level – and perhaps even be fun – because working together for a better future builds communities and helps to overcome social divisions. Let's hope that Lenton's heartening account will motivate many people to increase their efforts to address climate change.

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People in Venice, Italy, plant trees to protect crucial salt-marsh ecosystems.